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TITLE: Multi-Arm Protocol for Recognizing and Treating Alcohol Withdrawal in the ICU Patient

PROTOCOL: Alcohol Withdrawal Syndrome

PURPOSE: The purpose of this protocol is to outline criteria for early recognition of patients experiencing or at risk of alcohol withdrawal syndrome (AWS) and to provide a framework for selecting optimal treatment strategies.

APPROVED:



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DEFINITIONS:

- A. **Alcohol Withdrawal Syndrome (AWS)** is a neuroexcitatory state caused by the abrupt cessation of alcohol use in those with alcohol dependency. The Diagnostic and Statistical Manual of Mental Disorders (DSM-5) requires the fulfillment of four criteria listed below.¹ The symptoms of AWS include tremor, sweating, tachycardia, hypertension, headache, fever, agitation, hallucinations, and seizures. AWS is thought to be caused by physiologic adaptations in response to chronic alcohol use including downregulation of gamma-aminobutyric acid (GABA) receptors and upregulation of excitatory N-methyl-D-aspartate (NMDA) receptors.²
- i. Cessation or reduction in heavy and prolonged alcohol use
 - ii. Two or more symptoms of alcohol withdrawal within hours to days
 - iii. Distress or functional impairment
 - iv. Symptoms are not otherwise attributable
- B. **Alcohol Use Disorder (AUD)** is a clinical diagnosis of alcohol use that involves some degree of impairment or distress due to alcohol consumption. The DSM-5 lists three categories including impaired control, social dysfunction/physical risk, and physiological dependence of which the patient must meet at least two out of eleven criteria.¹
- C. **Alcohol dependence** and **alcohol abuse** are terms that were used previously and have been replaced with alcohol use disorder in the DSM-5.³
- D. **Delirium Tremens** is a severe form of alcohol withdrawal characterized by agitated delirium with or without generalized tonic-clonic seizures. This occurs in approximately 3-5% of patients hospitalized for alcohol withdrawal.⁴
- E. **High risk** for alcohol withdrawal syndrome^{5,8}
- a. Men: more than 4 drinks on any day or more than 14 drinks per week
 - b. Women: more than 3 drinks on any day or more than 7 drinks per week
 - c. History of alcohol withdrawal seizures or delirium tremens
 - d. GABA agonists including chronic benzodiazepine or barbiturate use
 - e. Alcohol Use Disorders Identification Test (AUDIT) score greater than or equal to 8
 - f. Initial alcohol level greater than 100 without signs of alcohol intoxication
- F. Scoring Tools
- a. The **Alcohol Use Disorders Identification Test (AUDIT)** score is a screening tool for AUD with scores > 4 indicating heavy drinking and scores > 8 correlating with withdrawal syndrome during the admission (100% sensitivity, 90% specificity).⁶ See appendix D for the AUDIT screening tool.

- b. **Clinical Institute Withdrawal Assessment (CIWA)** is a numerical score to grade the severity of alcohol withdrawal symptoms. This score has previously been used to guide benzodiazepine therapy for withdrawal and is currently recommended by the American Academy of Family Physicians (AAFP).⁷ Studies have demonstrated decreased treatment duration, length of stay, and seizure occurrence with symptom-based benzodiazepine regimens as compared to fixed dose.⁸ Despite the benefits of symptom-based regimens, the CIWA score is time consuming and requires patient participation/grading which is limited in patients undergoing severe alcohol withdrawal and patients who are intubated.
- c. **Richmond Agitation-Sedation Scale (RASS)** is a validated ICU score and is supported for monitoring symptoms by the American Society of Addiction Medicine for monitoring withdrawal in patients in the ICU.⁸ There is one retrospective study that demonstrated similar results to the CIWA score when used for symptom monitoring.⁹
- d. Shands-specific **alcohol withdrawal score (AWS)** is an institutional scoring system that divides symptoms into type A, B, and C categories and provides a numeric value to measure severity. These distinctions are not currently represented in societal guidelines or literature. See appendix E.

FIRST LINE TREATMENTS:

1. **Phenobarbital** is a barbiturate that has dual action by enhancing GABA inhibitory effects and blocking glutamate excitation.⁸ It has a long half-life of 80-120 hours.¹⁰ Weight based loading doses have demonstrated faster time to resolution of symptoms, decreased need for restraints, shorter length of stay, and reduced need for intubation compared to benzodiazepines.^{11,12} Dosing strategies include symptom based dosing, loading dose with taper, and loading dose without taper. Studies have shown that a loading dose without taper is noninferior to a loading dose with taper.¹³ The primary disadvantages of phenobarbital are that it is not reversible and that it has several clinically significant drug-drug interactions. Contra-indications to phenobarbital include porphyria, CYP enzyme-metabolized medications (HIV antiretrovirals, coumadin, immune suppressants, contraceptives, QT prolonging medications), and advanced cirrhosis (Child-Pugh C or hepatic encephalopathy).^{14,15} See appendix A for phenobarbital pathway.
2. **Benzodiazepines** have historically been the first line treatment for alcohol withdrawal and have been demonstrated in multiple studies to reduce the risk of withdrawal related seizures and delirium tremens.¹⁶ Lorazepam is a benzodiazepine with an intermediate half-life of 12-19 hours. Advantages over phenobarbital include fewer drug-drug interactions, less accumulation in patients with hepatic dysfunction, ability to titrate effect due to shorter half-life, and ability to reverse the medication. Disadvantages include the need for

more frequent, reactive dosing and the need for tapering.¹⁷ The lorazepam pathway is preferred for patients who are pregnant, though phenobarbital has similar teratogenic considerations.⁸ See appendix B for the lorazepam pathway.

3. See appendix C for adjunct treatments.

CORE PROCEDURE:

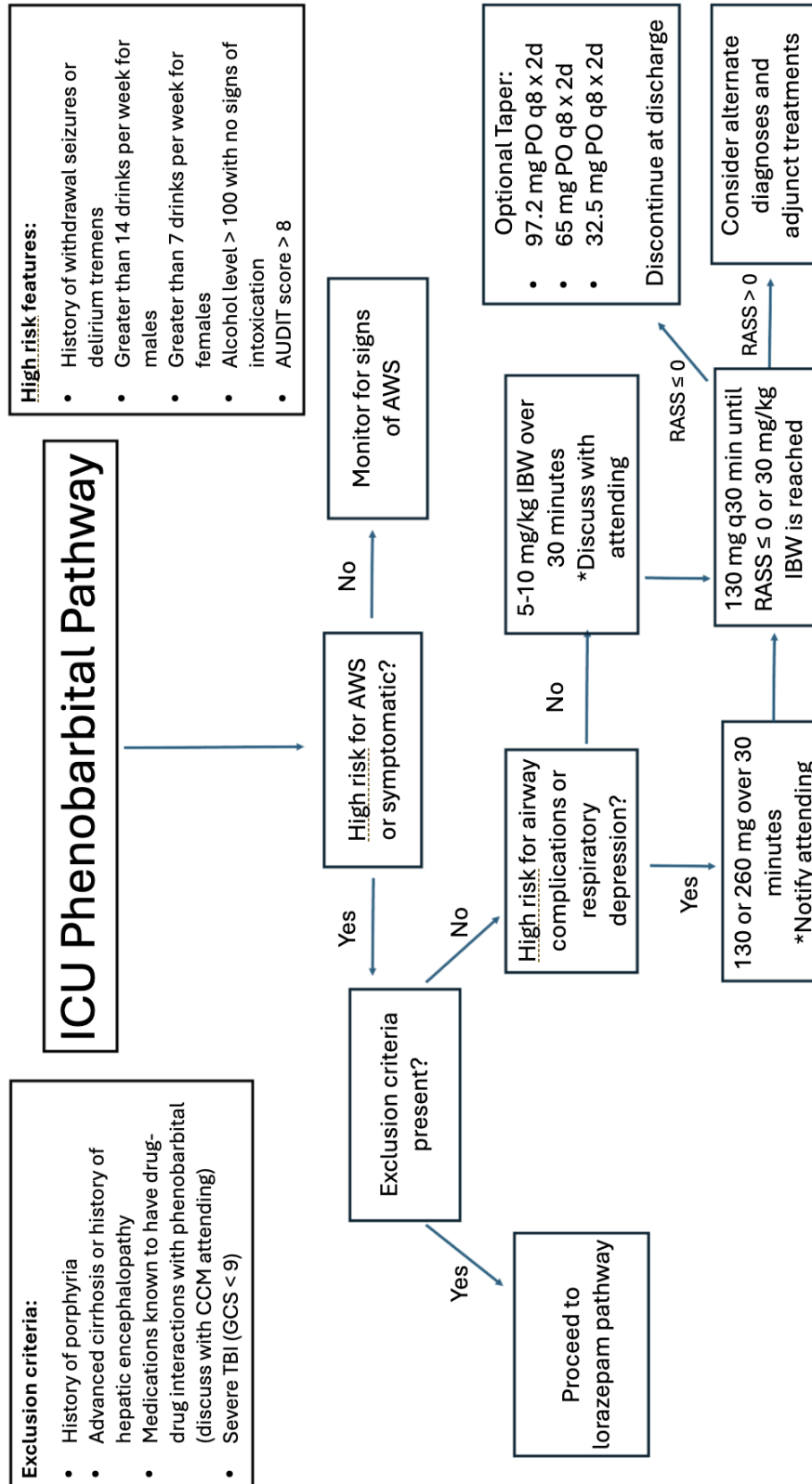
1. Determine whether the patient meets criteria for AWS prophylaxis or is experiencing symptoms of withdrawal requiring treatment.
2. Determine which pathway is clinically indicated for the patient (phenobarbital vs lorazepam) and follow the flow chart in the appendix with the therapeutic goal being RASS of -1 to 0.
 - a. If Phenobarbital pathway is chosen, CCM attending must be notified prior to administration
 - b. Phenobarbital pathway should NOT be used in patients with TBI
 - c. Phenobarbital pathway should NOT be used at night if AUDIT score does not designate as high risk or if unable to be obtained
3. If symptoms persist despite following the pathway, consider alternate diagnoses. If AWS is still suspected, consider the use of adjunct treatments.
4. Consider consultation with social work, addiction medicine, and geriatrics as clinically indicated.
5. If the patient is started on phenobarbital and is found to have an active outpatient prescription for HIV therapy, phenobarbital should immediately be discontinued, and an outpatient referral for follow up with infectious disease should be placed.

KEYWORDS:

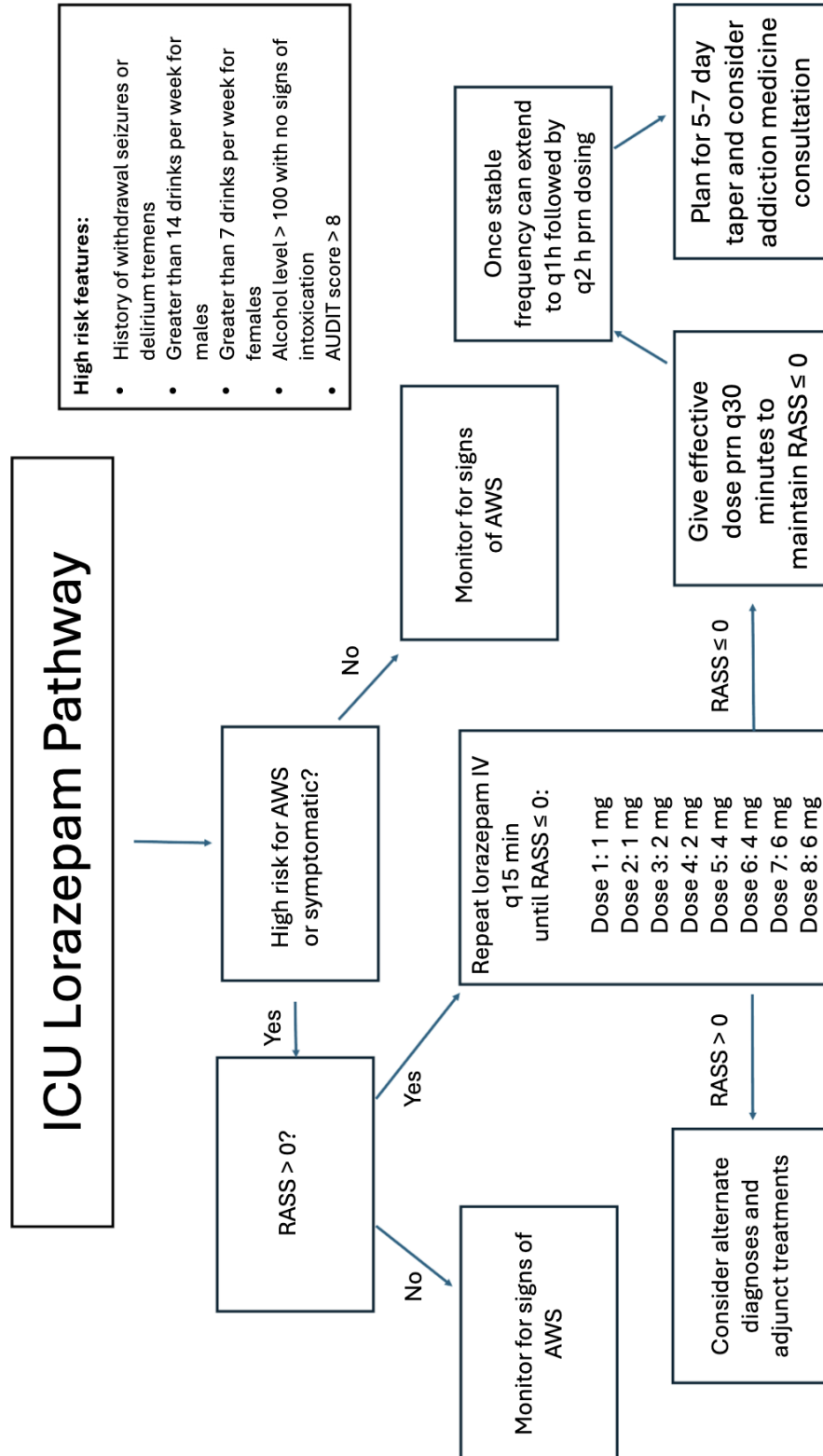
Alcohol withdrawal syndrome, phenobarbital, lorazepam, CIWA, RASS

APPENDICES

APPENDIX A – Phenobarbital Pathway



APPENDIX B – Lorazepam Pathway



APPENDIX C – Adjunct Therapies

*Only to be used in conjunction with phenobarbital or benzodiazepines; these medications should not be used as monotherapy

Medication	Mechanism	Typical Dosing	Contraindications/Considerations
Dexmedetomidine	α 2-adrenergic agonist	Continuous infusion: 0.2–0.7 mcg/kg/hr (titrate to effect)	Heart block, severe bradycardia; not for monotherapy (does not prevent seizures or delirium)
Propofol	GABA-A agonist	Continuous infusion: 5–80 mcg/kg/min (titrate to sedation goal)	Only for patients already intubated or when other adjuncts have failed
Clonidine	α 2-adrenergic agonist	0.1–0.2 mg PO q6–8h (max 0.8 mg/day)	Hypotension, bradycardia; not for monotherapy (does not prevent seizures or delirium)
Guanfacine	α 2-adrenergic agonist	No data	Animal studies have positive results, but no clinical data in humans
Gabapentin	Calcium channel modulation	300–600 mg PO q6–8h; up to 1,200 mg/day	Renal impairment (dose adjustment required); misuse potential; primary study was in outpatient population
Haloperidol	D2 receptor antagonist	0.5–5 mg IV/IM q4–8h PRN for hallucinations	Not for monotherapy (lowers seizure threshold); QTc prolongation, Parkinson's disease
Quetiapine Risperidone	Atypical antipsychotic	Quetiapine: 25–100 mg PO BID-TID; Risperidone: 0.5–2 mg PO BID	Not for monotherapy; caution in elderly (increased mortality risk), QTc prolongation
Carbamazepine	Sodium channel blockade	200 mg PO q8h or 400 mg PO q12h	Bone marrow suppression, hepatic impairment, Steven-Johnson syndrome risk
Valproate	GABA enhancement	300–500 mg PO/IV q6h	Pregnancy (teratogenic), liver disease; not recommended as monotherapy for AWS
Thiamine	Cofactor for carbohydrate metabolism	100 mg IV/IM daily for 3-5 days or 200 mg IV q8h for 3 days followed by 100 mg PO/IV daily	Give high dose for suspected Wernicke's encephalopathy; administer before or simultaneously if dextrose is needed
Folic acid	Coenzyme for DNA/RNA synthesis	Consider 1 mg PO daily	Weak recommendation by ASAM, which notes that vitamin deficiencies are expected to rapidly correct with restoration of diet

APPENDIX D – AUDIT Screening Tool (World Health Organization)

The Alcohol Use Disorders Identification Test: Interview Version Read questions as written. Record answers carefully. Begin the AUDIT by saying "Now I am going to ask you some questions about your use of alcoholic beverages during this past year." Explain what is meant by "alcoholic beverages" by using local examples of beer, wine, vodka, etc. Code answers in terms of "standard drinks". Place the correct answer number in the box at the right.	
1. How often do you have a drink containing alcohol? (0) Never [Skip to Qs 9-10] (1) Monthly or less (2) 2 to 4 times a month (3) 2 to 3 times a week (4) 4 or more times a week	6. How often during the last year have you needed a first drink in the morning to get yourself going after a heavy drinking session? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily
2. How many drinks containing alcohol do you have on a typical day when you are drinking? (0) 1 or 2 (1) 3 or 4 (2) 5 or 6 (3) 7, 8, or 9 (4) 10 or more	7. How often during the last year have you had a feeling of guilt or remorse after drinking? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily
3. How often do you have six or more drinks on one occasion? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily <i>Skip to Questions 9 and 10 if Total Score for Questions 2 and 3 = 0</i>	8. How often during the last year have you been unable to remember what happened the night before because you had been drinking? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily
4. How often during the last year have you found that you were not able to stop drinking once you had started? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily	9. Have you or someone else been injured as a result of your drinking? (0) No (2) Yes, but not in the last year (4) Yes, during the last year
5. How often during the last year have you failed to do what was normally expected from you because of drinking? (0) Never (1) Less than monthly (2) Monthly (3) Weekly (4) Daily or almost daily	10. Has a relative or friend or a doctor or another health worker been concerned about your drinking or suggested you cut down? (0) No (2) Yes, but not in the last year (4) Yes, during the last year
Record total of specific items here <i>If total is greater than recommended cut-off, consult User's Manual.</i>	

Appendix E – University of Florida AWS score

UF Health Shands AWS Scoring System	
Assign 1 point for each symptom	
Type A Symptoms (CNS Excitation) Does Patient appear:	
a. Anxious or nervous	
b. Restless	
c. Bothered by bright light	
d. Bothered by loud sounds	
Type B Symptoms (Adrenergic Hyperactivity)	
e. Nausea and/or vomiting	
f. Visible tremor with or without arms extended	
g. Visible sweat on palms or forehead	
h. Systolic blood pressure greater than 160 mm Hg	
i. Diastolic blood pressure greater than 100 mm Hg	
j. Heart rate greater than 100 beats per minute	
k. Extra or skipped beats on apical pulse	
Type C Symptoms (Delirium) Does the patient:	
l. Respond inappropriately to questions	
m. Report hearing noises that are not there	
n. Report seeing something that is not there	
o. States own name incorrectly	
p. Answer where they are incorrectly	
q. Answer how long they have been hospitalized incorrectly	
r. Answer the year incorrectly	
s. Answer the month incorrectly	
t. Answer the day of the week incorrectly	

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